NET731 – Networking

Week 7 Practical: Monitoring Setup

17–23 April | Total Marks: 10

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Submission Date:	15/05/2026
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Learning Outcomes

By the end of this practical, you should be able to:

- Enable and configure Azure Monitor on virtual machine and network resources
- Navigate the Azure Monitor interface to locate and interpret key metrics
- Identify normal versus abnormal performance baselines for CPU, network, and disk
- Produce a structured monitoring report with supporting evidence

Overview

This week you will configure Azure Monitor across the cloud environment you have been building throughout the semester. Azure Monitor is the central platform for collecting, analysing, and acting on telemetry from your Azure resources. Effective monitoring is not simply about enabling tools — it requires understanding what you are measuring and why it matters.

You are expected to make decisions, explore the interface, and think critically about what you observe. Screenshots must clearly show your own Azure environment — generic or placeholder images will not be accepted.

Before You Begin — Prerequisites

Ensure your VM from previous weeks is running and accessible.
Your Virtual Network (VNet) and Network Security Group (NSG) from Week 5 should be in place.
You will need Contributor access to your Azure resource group.
Allow 10–15 minutes for monitoring data to populate after enabling Azure Monitor.

Task 1 — Enable Azure Monitor (4 Marks)

Azure Monitor needs to be enabled on individual resources before it can collect data. This task covers both your virtual machine and the network resources in your environment.

Task 1.1 — Enable Monitoring on Your Virtual Machine (2 Marks)

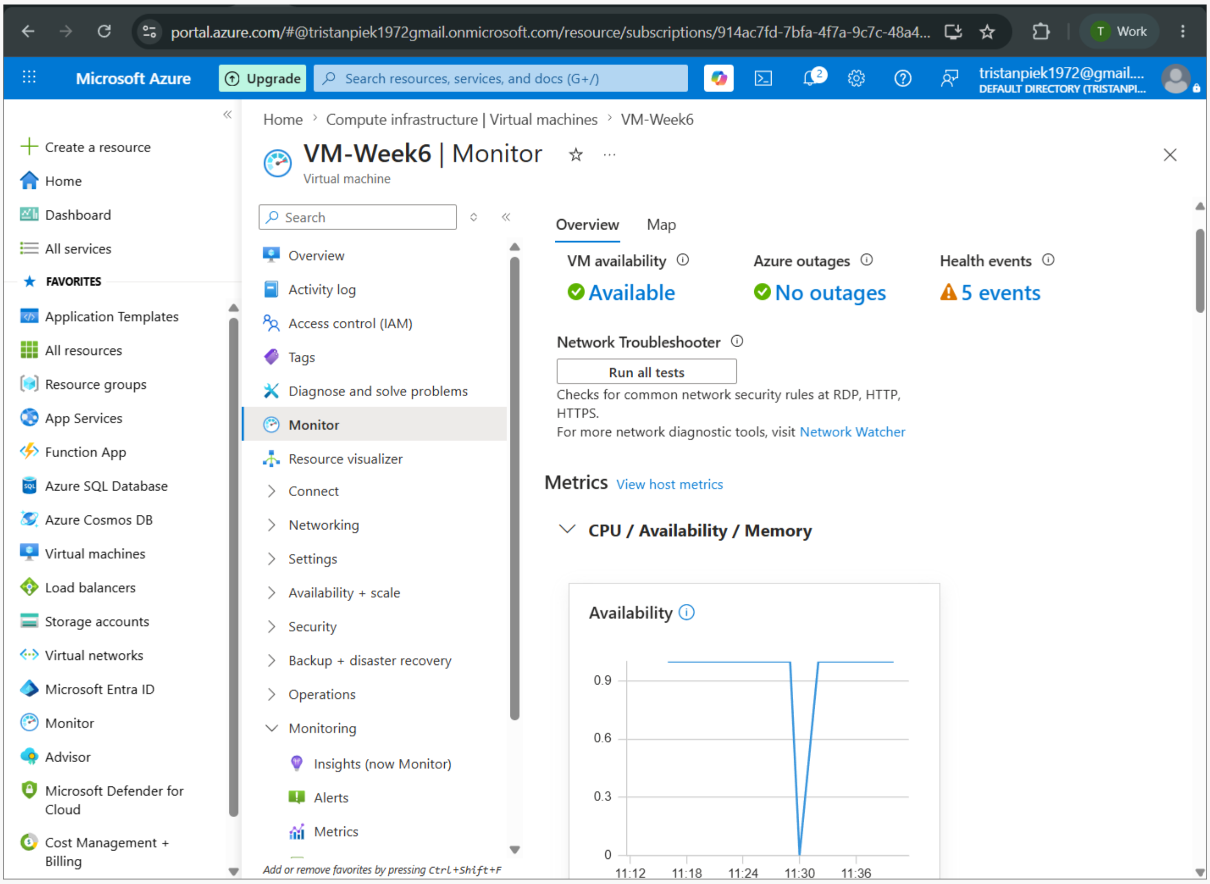
Navigate to your virtual machine in the Azure Portal and locate the Monitoring section in the left-hand menu. You will find several sub-options here — explore them before enabling anything.

Think Before You Click

What is the difference between 'Insights' and 'Diagnostics settings' for a VM?
What data will be collected once you enable monitoring? Where is it stored?
Does enabling monitoring have any cost implications you should be aware of?

Once you have explored the options, enable VM Insights or Diagnostics for your virtual machine. Configure it to collect at minimum: CPU metrics, memory (if supported), and disk I/O.

[Screenshot: Azure Monitor enabled on VM — show the Monitoring blade confirming activation]



Describe in 2–3 sentences what you enabled and where the collected data will be stored:

Your response: I enabled Azure Monitor Insights and Diagnostic Settings to monitor the virtual machines health and performance. The Analytics Workspace will keep monitoring data for disc, memory, and CPU metrics across time.

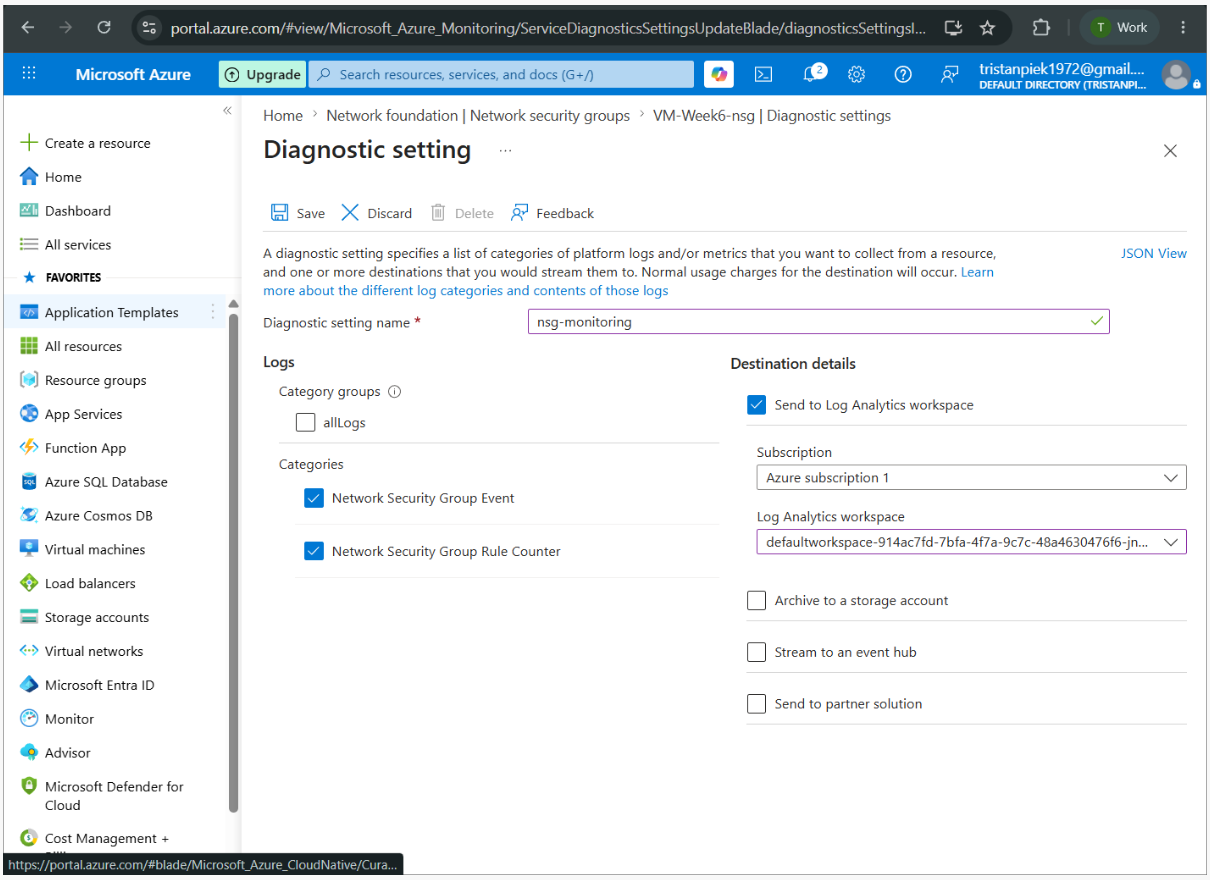
Task 1.2 — Enable Monitoring on Network Resources (2 Marks)

Repeat the process for your network resources. Both your NSG and your VNet support diagnostic settings.

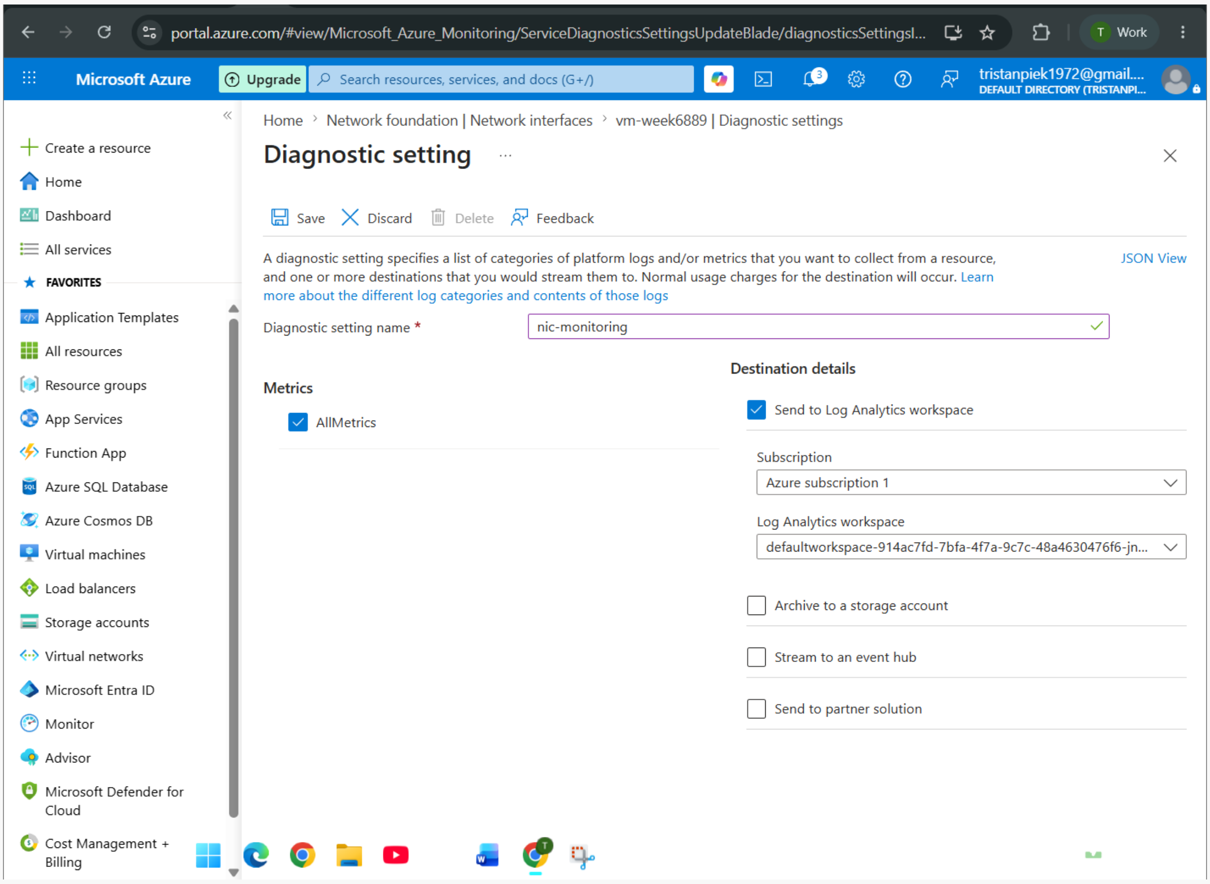
Consider the following before proceeding:

- What specific log categories are available for an NSG? Which ones would be most useful in a security context?
- What is a Log Analytics Workspace, and why might you need one to proceed?
- If you do not already have a Log Analytics Workspace, you will need to create one — consider which region is appropriate.

[Screenshot: Diagnostic settings enabled on NSG — show categories selected and destination]



[Screenshot: Diagnostic settings on VNet or additional network resource]



Task 2 — Review Key Metrics (3 Marks)

Now that monitoring is enabled, you will use Azure Monitor's Metrics Explorer to observe and interpret the data being collected. Generating some load on your VM before taking screenshots will produce more meaningful graphs.

Generating Test Load (Optional but Recommended)

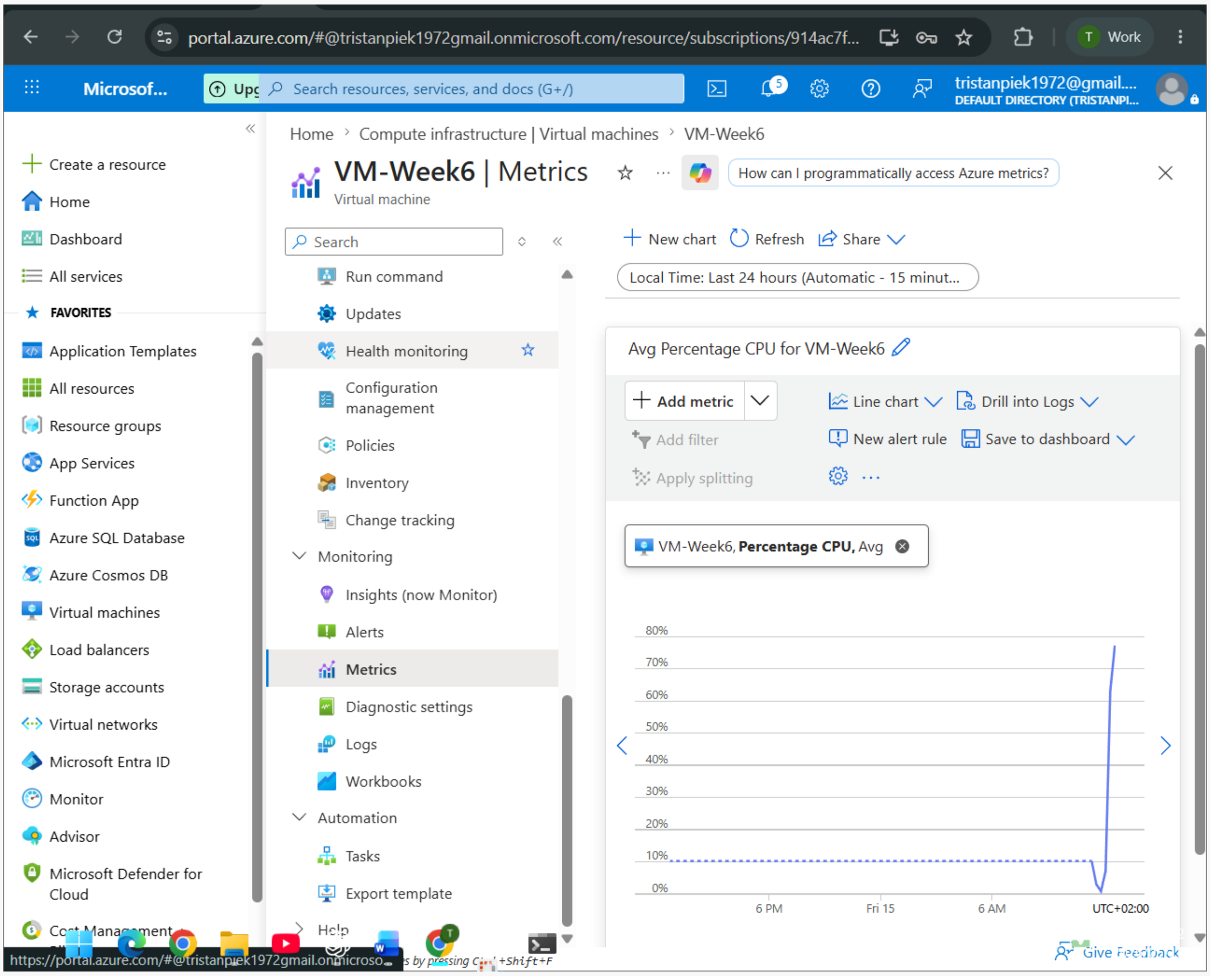
On Linux VMs: the 'stress' tool can simulate CPU and I/O load.
On Windows VMs: Task Manager's 'New Task' can run a CPU-intensive process.
Even browsing the web or installing packages from within the VM will generate measurable network activity.
Allow a few minutes after generating load before capturing screenshots.

Task 2.1 — CPU Metrics (1 Mark)

Open Azure Monitor > Metrics and select your VM as the scope. Explore the available metrics before choosing one to display.

- Select an appropriate time range that shows meaningful data (consider: what time range makes sense for a lab exercise?)
- Identify which metric reflects actual CPU usage — consider why there might be more than one CPU-related metric
- Add annotations or use the pin feature to highlight any interesting patterns you observe

[Screenshot: CPU metrics chart — VM scope, showing usage over a meaningful time period]



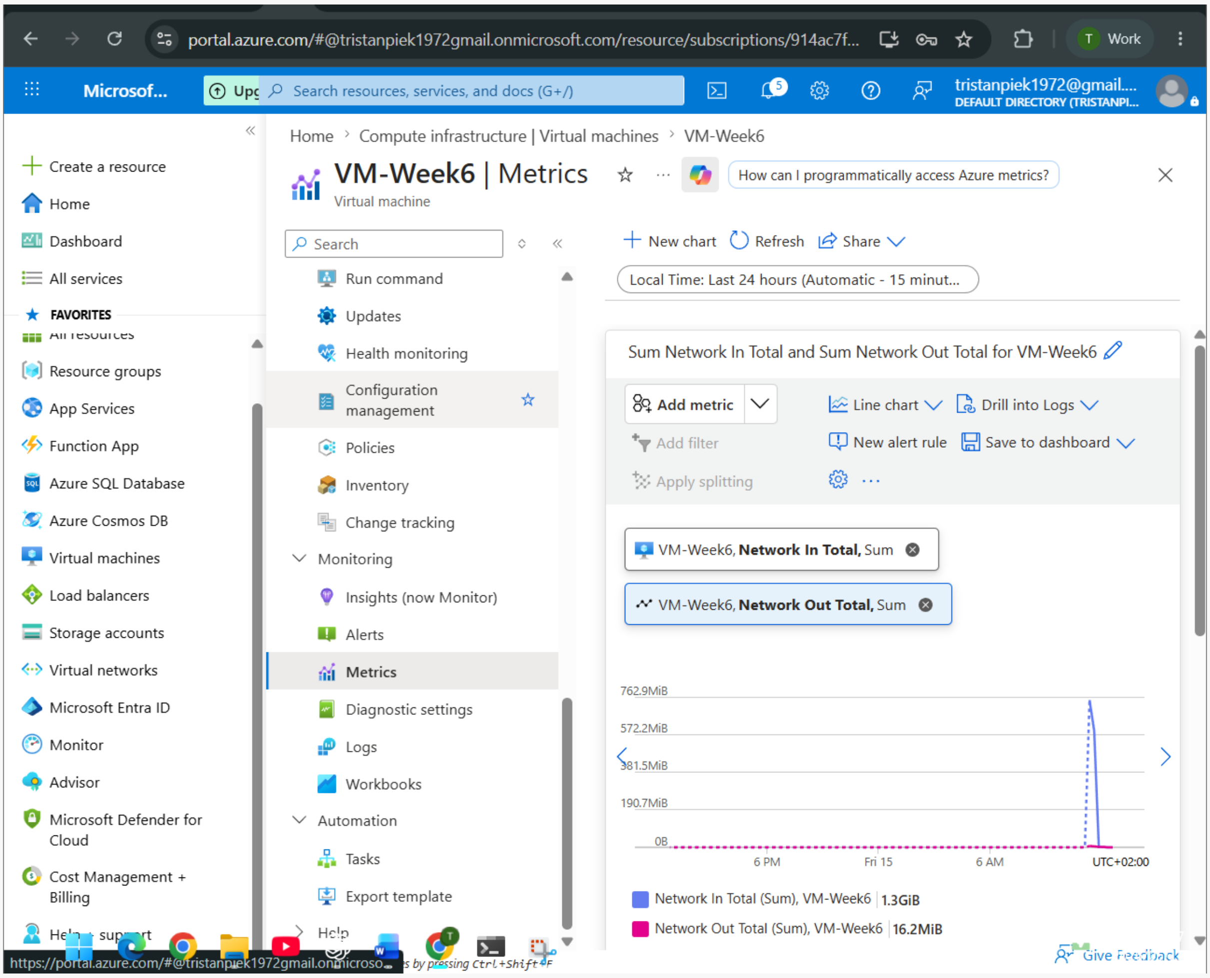
In 1–2 sentences, describe what the CPU metric is showing and whether the values appear expected for your workload: The CPU statistic show VM processor usage % over time. Values increased during workload generation simulating activity with CPU intensive jobs.

Task 2.2 — Network Metrics (1 Mark)

Still in Metrics Explorer, change the metric to observe network throughput. You should investigate both inbound and outbound traffic.

- What is the unit of measurement for the network metric you selected? Is it what you expected?
- Can you display both Network In and Network Out on the same chart? How would you do this?

[Screenshot: Network metrics chart — showing inbound and/or outbound traffic]



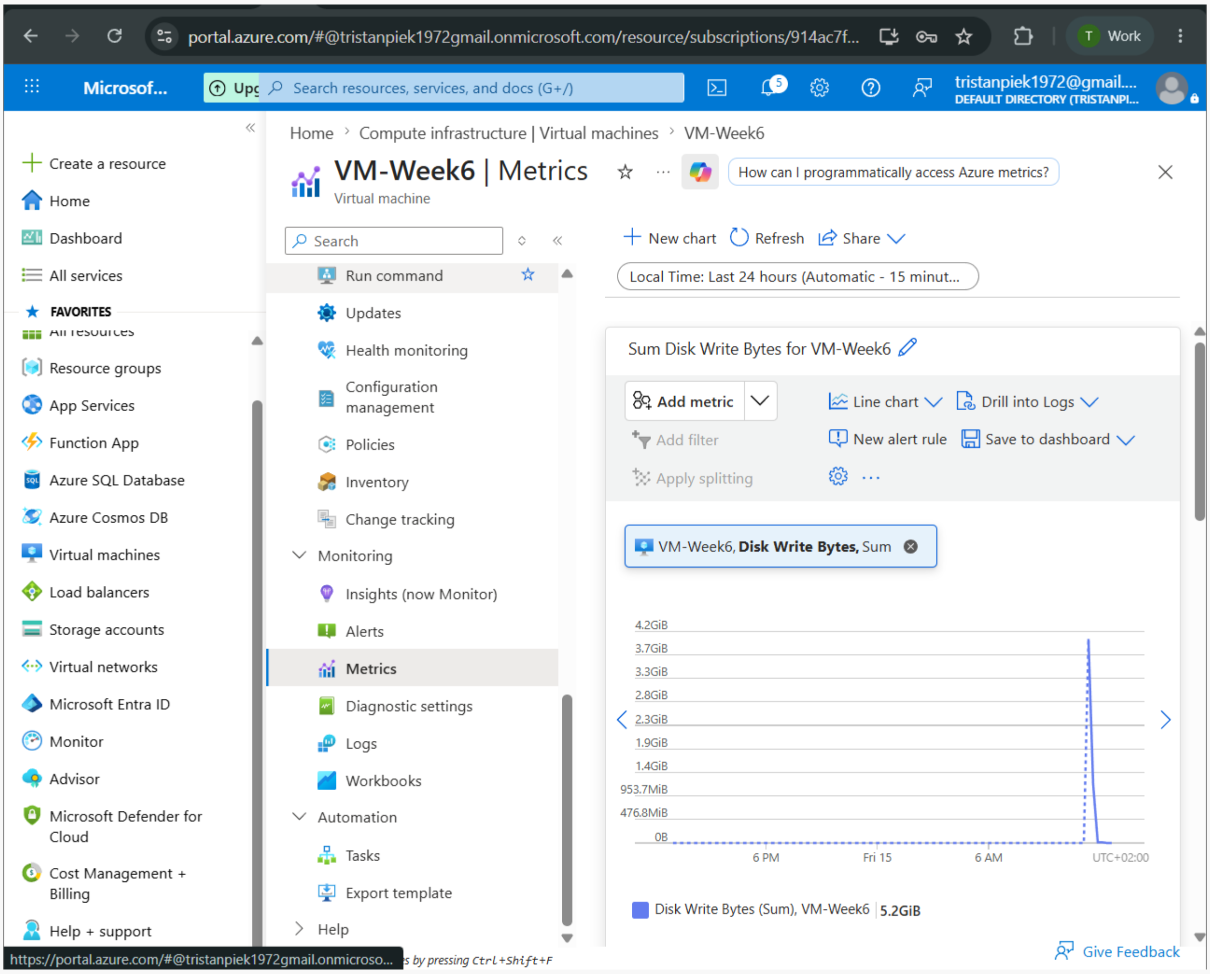
Note any observations about the network traffic pattern you can see: These two charts show all Inbound and Outbound traffic going through the VN interface. The chart indicates that monitoring is active and successful and is capturing data.

Task 2.3 — Disk Metrics (1 Mark)

Finally, explore the disk-related metrics for your VM. Azure exposes separate metrics for disk reads and disk writes.

- Locate both Disk Read Bytes and Disk Write Bytes (or equivalent metrics in your VM configuration)
- Consider: during what activities would you expect disk write activity to be high?

[Screenshot: Disk metrics chart — showing read and/or write activity]



Describe in 1–2 sentences what the disk metrics reveal about your VM's activity: VM storage disc metrics reflect read and write operations. As expected in workload simulation, file creation and system operations increased disc write activity.

Task 3 — Document Your Monitoring Setup (3 Marks)

Monitoring documentation is a critical deliverable in professional environments. A future administrator — or your future self — should be able to understand your setup from the document alone.

Task 3.1 — Monitoring Summary (2 Marks)

Write a structured summary of your monitoring configuration. Your summary should cover all three areas below. Aim for clarity and precision — this is professional technical documentation.

Resources Monitored

List each Azure resource you have enabled monitoring on, specifying the monitoring method used (e.g., VM Insights, Diagnostic Settings) and what data each resource is collecting:

The virtual machine and network resources such as the NSG, were set to be monitored. Azure Monitor was employed to gather data to server utilisation, disc activity, network traffic, and security related happenings. A Log analytics workspace was integrated with the data, allowing for centralised viewing and analysis.

Data Retention and Storage

Where is your monitoring data stored? What retention period applies? How would you access historical data older than the default retention window?

An Log analytics workspace is where all of the monitoring data is saved. If issues arise in the future, this implies that diagnostic and performance data is stored for a while. When you need to save logs for a long time you can also archive them.

Metric Observations

Summaries of what you observed across the three metrics reviewed in Task 2. Were any values unexpected? What might cause anomalies in these metrics in a production environment?

When workloads were generated on the virtual machine during testing, the CPU utilisation increased. Creating processes inside the machine and visiting the web also boosted network traffic. When files were being created, disc write activity could be seen. System problems, severe workload and security concerns could be indicated by large or unexpected spikes in these measurements.

Task 3.2 — Extension: Configure an Alert Rule (1 Bonus Mark)

Extension Task

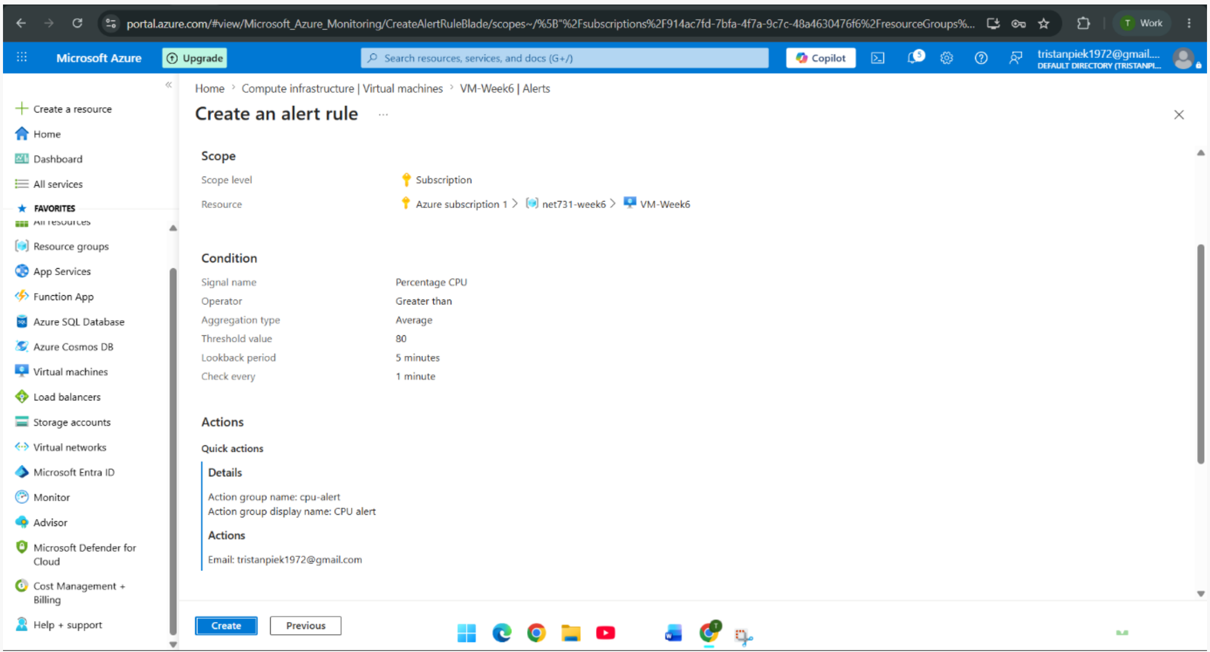
This task is optional but provides an opportunity to demonstrate deeper understanding.

Configure at least one Azure Monitor Alert Rule on your VM or network resource.

The alert should have a meaningful threshold — for example, triggering when CPU exceeds 80% for 5 minutes.

Document your alert logic: what condition triggers it, what action does it take, and why you chose that threshold.

[Screenshot: Configured alert rule — show condition, threshold, and action group]



Explain the alert rule you configured and the reasoning behind your chosen threshold: I set an Azure Monitor alert rule for 5 minutes of CPU utilisation over 80%. This level was chosen because high CPU consumption may occur causing resource fatigue or unexpected application behaviour.

Reflection Questions

Answer these questions thoughtfully. There are no single correct answers — markers are looking for evidence of understanding and critical thinking.

Reflection Question	Your Response
Why is monitoring important in a cloud environment? How does it differ from monitoring on-premises infrastructure?	Integrated monitoring tools are an essential component of cloud computing platforms such as Azure. These tools enable the automated collecting and analysis of data across a wide range of applications. On premises monitoring often requires a greater amount of human engagement for both the initial setup and ongoing maintenance purposes.
What is the difference between a metric and a log in Azure Monitor? When would you use each?	Metrics like CPU use and network traffic measure system performance over time. Logs detail system events and activity. Using monitoring and alerting are

	faster with metrics, while troubleshooting and investigation are better with logs.
If your VM showed consistently high CPU usage (above 90%) every day at 2am, what steps would you take to investigate?	See if there are no scheduled jobs, updates, or backups. I would also check statistics and logs to find the CPU-hogging process. If activity seemed strange, I would check for security or performance issues.
How could monitoring data be used to make cost-optimisation decisions in Azure?	The monitoring data can tell us if we are overusing or underusing resources. If a VM is using very little CPU or memory for the most part it could be downsized to a smaller and cheaper option. Monitoring can also reveal opportunities to reduce costs by reducing resources.

Marking Criteria

Use this table to check your submission before submitting.

Task	Marks	Evidence Required
1.1 Enable Azure Monitor on VM	**2**	*Screenshot of Monitor enabled on VM resource*
1.2 Enable Monitoring on Network Resources	**2**	*Screenshot showing NSG/VNet diagnostics*
2.1 CPU Metrics Review	**1**	*Screenshot of CPU usage chart with annotations*
2.2 Network Metrics Review	**1**	*Screenshot of network in/out metrics*
2.3 Disk Metrics Review	**1**	*Screenshot of disk read/write metrics*
3.1 Monitoring Documentation	**2**	*Written summary with all screenshots included*
3.2 Alert Configuration (Extension)	**1**	*Screenshot of configured alert rule*
TOTAL	**10**	

Submission Checklist

All screenshot placeholders have been replaced with real screenshots from your Azure environment.
Each screenshot clearly shows your resource names and Azure subscription context.
Written responses are your own words — not copied from documentation.
Reflection questions have been answered with evidence of understanding.
The document has been saved as a .docx or .pdf and submitted via the course portal.

Useful References

The following official documentation may help — but always explore the Azure Portal first before consulting documentation:

- Azure Monitor overview — docs.microsoft.com/azure/azure-monitor/overview
- Enable VM Insights — docs.microsoft.com/azure/azure-monitor/vm/vminsights-enable-overview
- Metrics Explorer — docs.microsoft.com/azure/azure-monitor/essentials/metrics-getting-started
- Diagnostic settings — docs.microsoft.com/azure/azure-monitor/essentials/diagnostic-settings
- Azure Monitor Alerts — docs.microsoft.com/azure/azure-monitor/alerts/alerts-overview

Tip: The Azure Portal's built-in 'Learn more' links on each blade are often the fastest route to relevant documentation.